Growing Hypergraphs with Node Labels

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We study a generative mechanistic model of hypergraph growth in which nodes possess binary labels which generalizes prior work on edge-copying growing hypergraphs. Edges form in this model as noisy copies of previous edges, where the transmission of nodes from one edge to the next depends on the labels. We derive a power law for the degree distribution in this model and describe the long-term dynamics of the joint distribution of labels contained in edges. We then demonstrate that this model can be learned from data: we can both estimate the parameters of the model via (stochastic) expectation maximzation and perform community detection of the node labels via maximum-likelihood estimation.

I. INTRODUCTION

A. Higher order!!!

[1–3].
General hypergraph data mining:
[4]

B. Structural Properties of Higher-Order Network Data

[5]
[6]
Impact of overlap on dynamics: [7]

C. Generative Hypergraph Models

Higher-order intersections: [8]
CRU: [9]
Edge-copy model: [10]
THyME: [11]
A model based on diversity: [12]

D. Generative Models for Inference and Community Detection

[13] [14–16]
[17]

E. Need to categorize

[18]

F. Tooling

PyTorch: [19]
Simulated annealing? [20]
XGI!! [21]

II. LONG-RUN DYNAMICS

A. Joint Distribution of Edge Labels

We consider the joint distribution $p(k_0, k_1)$ of label counts in a uniformly random edge, where k_0 is the number of nodes with label 0 in the edge and k_1 is the number of nodes with label 1 in the edge. Although we are unable to find a closed-form expression for this distribution at stationarity, we can write down a linear map whose leading eigenvector corresponds to the stationary distribution. Let $p^{(t)}(k_0, k_1)$ be the joint distribution of label counts in a uniformly random edge after t timesteps. Then, the probability $q(k_0', k_1')$ of realizing an edge with k_0' nodes of label 0 and k_1' nodes of label 1 at time $t+1$ is given by

$$q(k_0', k_1') = \sum_{k_0,k_1} T(k_0', k_1' | k_0, k_1) p^{(t)}(k_0, k_1) \;, \qquad (1)$$

where $T(k_0', k_1' | k_0, k_1)$ gives the probability that the edge e realized in step $t+1$ has k_0' nodes of label 0 and k_1' nodes of label 1 given that the edge f sampled to form e has k_0 nodes of label 0 and k_1 nodes of label 1. At stationarity, we must have $q(k_0', k_1') = p^{(t)}(k_0', k_1') = p^{(t+1)}(k_0', k_1')$, so the stationarity condition can be obtained by solving the linear system

$$p(k_0', k_1') = \sum_{k_0,k_1} T(k_0', k_1' | k_0, k_1) p(k_0, k_1) \;. \qquad (2)$$

The stationary joint distribution $p(k_0, k_1)$ is therefore given by the leading eigenvector of the doubly-indexed matrix T with entries $T(k_0', k_1' | k_0, k_1)$. The entries of this matrix can be computed analytically. Conditional on choosing an edge f with k_0 nodes of label 0 and k_1

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nodes of label 1, a node of label z is selected as the seed node u with probability $k_z/(k_0+k_1)$. Then, the number of nodes of labels z and $\bar{z}$ in e have distributions

$$k'_z \sim 1 + \mathrm{Binom}\left(k_z - 1, \eta_+\right) + \mathrm{Poisson}(\lambda_+ + \gamma_+)\ ,$$
$$k'_{\bar{z}} \sim \mathrm{Binom}\left(k_{\bar{z}}, \eta_-\right) + \mathrm{Poisson}(\lambda_- + \gamma_-)\ .$$

The probability of obtaining k'_0 nodes of label 0, conditioning on k_0 and k_1, is

$$t(k'_0|k_0,k_1) = \frac{k_0}{k_0+k_1}\sum_{i=0}^{k'_0-1}\binom{k_0-1}{i}\eta_+^i(1-\eta_+)^{k_0-1-i}e^{-(\lambda_++\gamma_+)}\frac{(\lambda_++\gamma_+)^{k'_0-1-i}}{(k'_0-1-i)!}+ \tag{3}$$

$$\frac{k_1}{k_0+k_1}\sum_{i=0}^{k'_0}\binom{k_0}{i}\eta_-^i(1-\eta_-)^{k_0-i}e^{-(\lambda_-+\gamma_-)}\frac{(\lambda_-+\gamma_-)^{k'_0-i}}{(k'_0-i)!}\ . \tag{4}$$

A parallel expression describes $t(k'_1|k_0,k_1)$, and the transition matrix entries are given by the product

$$T(k'_0,k'_1|k_0,k_1) = t(k'_0|k_0,k_1)t(k'_1|k_0,k_1)\ . \tag{5}$$

In Figure 2, we show projections of the joint distribution $p^{(t)}(k_0,k_1)$ onto the top three eigenvectors of the matrix T during a simulation run. These eigenvectors are computed numerically. The Perron eigenvector v_1 with eigenvalue 1 corresponds to the stationary distribution $p(k_0,k_1)$, while the second and third eigenvectors v_2 and v_3 correspond to transient dynamics. The corresponding eigenvalues are relatively large, indicating that these transients decay relatively slowly.

B. Mean Degree

The mean degree $\langle d\rangle$ of a node satisfies

$$\langle d\rangle = \frac{m}{n}\langle k\rangle\ , \tag{6}$$

where $\langle k\rangle$ is the mean edge size, m is the number of edges, and n is the number of nodes. In expectation, there are $\gamma_- + \gamma_+$ new nodes added per new edge, which means that $\frac{m}{n} \to \frac{1}{\gamma_-+\gamma_+}$ as $m,n\to\infty$ in expectation. So, we have

$$\langle d\rangle = \frac{\langle k\rangle}{\gamma_-+\gamma_+}\ . \tag{7}$$

We don't have a closed-form expression for the stationary value of $\langle k\rangle$, but we can compute it numerically from the stationary distribution of the joint label counts $p(k_0,k_1)$.

C. Power Law Degree Distribution

This argument follows that of [10], which in turn is based on a derivation by [22].

Under the mean-field assumption that the degree of a node is independent of the ratio $\frac{k_0}{k}$ of the edge it is contained in, we can write down an argument to show that the degree distribution follows a power law and give a formula for the exponent. Define the moments

$$\rho_0 = \left\langle\frac{k_0^2}{k}\right\rangle\quad,\quad \rho_1 = \left\langle\frac{k_1^2}{k}\right\rangle\quad \text{and}\quad \rho_{01} = \left\langle\frac{k_0k_1}{k}\right\rangle\ . \tag{8}$$

Then, the expected number of nodes which are sampled in the edge-sampling step is

$$\sigma \triangleq 1 + \eta_+\left[\rho_0+\rho_1-1\right] + \eta_-\rho_{01}\ . \tag{9}$$

Importantly, σ depends only on the parameters and the moments of the joint distribution $p(k_0,k_1)$, which we are going to show how to get from linear algebra later. Let $\tilde{\sigma} = \frac{\sigma}{\langle d\rangle}$, where $\langle d\rangle$ is the average degree of a node in the hypergraph.

Under a mean-field assumption in which the degree d of a node is independent from the counts k_0 and k_1 of labels in an edge containing that node, we can give the expected change in the count of degree d nodes as

$$\tilde{\sigma}((d-1)p_{d-1}^{(t)} - dp_d^{(t)}) \tag{10}$$

Similarly, the expected number of nodes which are sampled in the extant node addition step is $\lambda_+ + \lambda_-$. This gives, in expectation for the change in degree d nodes,

$$n^{(t+1)}p_d^{(t+1)} - n^{(t)}p_d^{(t)} = \tilde{\sigma}\left[(d-1)p_{d-1}^{(t)} - dp_d^{(t)}\right] + (\lambda_++\lambda_-)(p_{d-1}^{(t)} - p_d^{(t)})\ . \tag{11}$$

We know that at stationarity, $n^{(t+1)} - n^{(t)}$ is in expectation equal to $\gamma_+ + \gamma_-$, while at stationarity $p_d^{(t+1)} = p_d^{(t)} = p_d$. Thus, we obtain

$$p_d\,(\gamma_+ + \gamma_-) = \tilde{\sigma}\,[(d-1)p_{d-1} - dp_d] + (\lambda_+ + \lambda_-)(p_{d-1} - p_d)\,. \tag{12}$$

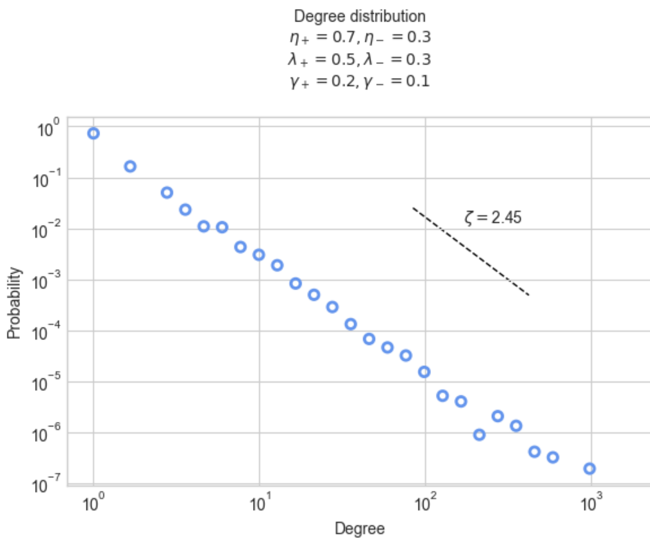

FIG. 1. Degree distribution from simulation (points) and theoretical power law (line) with exponent computed from model parameters and joint label distribution moments. PC: Put three series on this plot and give slopes for each. Also, fairly sure the calculation is bugged in either math or code.

Rearranging, we obtain

$$\frac{p_d}{p_{d-1}} = \frac{\tilde{\sigma}(d-1) + \lambda_+ + \lambda_-}{\tilde{\sigma}d + \gamma_+ + \gamma_- + \lambda_+ + \lambda_-} \tag{13}$$

$$= 1 - \frac{\tilde{\sigma} + \gamma_+ + \gamma_-}{\tilde{\sigma}d + \gamma_+ + \gamma_- + \lambda_+ + \lambda_-} \tag{14}$$

Allowing d to become large (which describes the tail of the degree distribution), we have

$$\frac{p_d}{p_{d-1}} \approx 1 - \frac{\tilde{\sigma} + \gamma_+ + \gamma_-}{\tilde{\sigma}d}\,, \tag{15}$$

a relation which corresponds to a power law with exponent $\zeta = 1 + \frac{\gamma_+ + \gamma_0}{\tilde{\sigma}}$. Explicitly calculating this moment requires computing the moments ρ_0, ρ_1, and ρ_{01}, which can be done computationally from the stationary joint distribution $p(k_0, k_1)$.

D. Joint Distribution of Labels

III. MODEL INFERENCE

PC: Someone needs to establish standardized notation in this section.

A. Parameter Estimation via Stochastic EM

PC: Violet, this is your spot! Descriptions of methods/algorithms should go here, results are for later.

Stochastic [23] expectation-maximization [24].

B. Community Detection via MLE

PC: Frannie, this is your spot! Descriptions of methods/algorithms should go here, results are for later.

IV. INFERENCE RESULTS

How did we do on real and synthetic data?

V. CONCLUSION

Blah blah blah.

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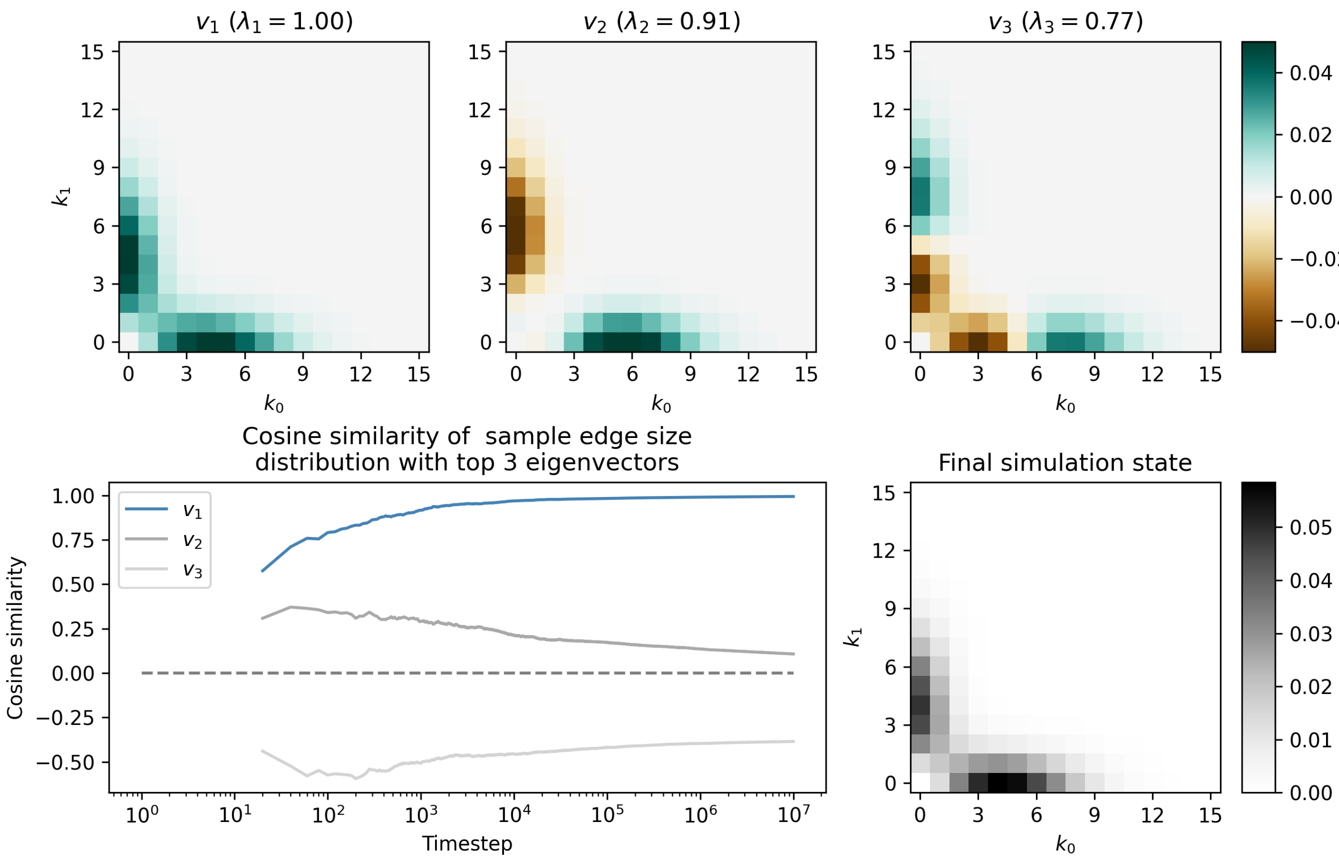

FIG. 2. Dynamics! Top shows the three leading eigenvectors of the distribution $p(k_0, k_1)$. Bottom left gives the cosine similarity (normalized projection) between one run of simulation dynamics and these eigenvectors. Bottom right gives the final simulation state. PC: Need to check these projections, looks a bit strange. Can run 10^7 steps in two minutes on laptop.

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